Program : **Diploma in Fire Technology and Safety**	
Course Code: **2461**	Course Title: **Principles of Safety Management**
Semester: **2**	Credits: **3**
Course Category: **Engineering Science**	
Periods per week: **3 (L: 3 T:0 P:0)**	Periods per semester: **45**

Course Objectives:

Foundationof Safety Engineering: To provide a comprehensive understanding of safety engineering principles and the importance of safety in various environments.

Accident Analysis and Prevention: To equip students with the ability to analyze accident causation theories and implement effective prevention methods.

Safety Policies and Training Evaluation: To develop skills in assessing the effectiveness of safety policies and training programs within organizations.

PPE Knowledge: To familiarize students with personal protective equipment (PPE) standards and their practical application in workplace safety.

Course Prerequisites:

Topic	Course code	Course Name	Semester
Basic knowledge in Physics	1003	Applied Physics I	1

Course Outcomes:

On completion of the course, the students will be able to:-

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand the principles of safety engineering and the importance of safety in various environments.	10	Applying
CO2	Analyze accident causation theories and apply prevention methods.	11	Understanding
CO3	Assess the effectiveness of safety policies and training programs in the workplace.	11	Understanding

CO4	Conduct accident investigations and analyze safety performance metrics.	11	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2			3		
CO2	2	2					
CO3	2				2		
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	**Understand the principles of safety engineering and the importance of safety in various environments.**		
M1.01	Define key safety terms and concepts related to safety engineering.	2	Understanding
M1.02	Explain the context of the safety movement and its significance.	3	Understanding
M1.03	Identify various stakeholders' roles in promoting safety within an organization.	3	Understanding
M1.04	Analyze different theories of accident causation and their implications for safety practices	2	Applying

Contents:

Introduction: Safety - Goals of safety engineering, Need for safety, Safety and productivity

Definitions: Accident, Injury, Unsafe act, Unsafe condition, Dangerous occurrence, Reportable accidents

History of Safety Movement: Overview of the evolution of safety practices

Theories of Accident Causation

Safety Organization: Objectives, types, functions - Role of management, supervisors, workmen, unions, government, and voluntary agencies in safety -Safety policy - Safety Office: responsibilities and authority -Safety committee : need, types, and advantages

CO2	**Analyze accident causation theories and apply prevention methods.**		
M2.01	Evaluate different methods of accident prevention in the workplace.	4	Understanding
M2.02	Assess the importance of safety education and training for employees.	2	Understanding
M2.03	Identify barriers to effective communication in safety protocols.	3	Understanding
M2.04	Implement best practices for housekeeping using the 5S methodology.	2	Understanding
	Series Test – I	1	

Contents:

Accident Prevention Methods: Engineering, Education, and Enforcement

Safety Education & Training: Importance, Various training methods, Effectiveness of training, Behavior-oriented training

Communication: Purpose, Barriers to communication

Housekeeping: Responsibility of management and employees, Advantages of good housekeeping, 5S of housekeeping

Work Permit System: Objectives, Hot work and cold work permits, Typical industrial models and methodology, Entry into confined spaces

CO3	**Assess the effectiveness of safety policies and training programs in the workplace**.

M3.01	Identify various types of personal protective equipment (PPE) and their applications.	3	Understanding
M3.02	Analyze PPE standards and their importance in workplace safety.	3	Understanding
M3.03	Calculate and interpret safety performance metrics (frequency, severity, incidence, activity rates).	2	Understanding
M3.04	Conduct safety inspections and job safety analysis (JSA) to identify potential hazards.	3	Understanding

Contents:

Personal Protection in the Work Environment: Types of PPEs, Personal protective equipment : respiratory and non-respiratory equipment, Standards related to PPEs

Monitoring Safety Performance: Frequency rate, Severity rate, Incidence rate, Activity rate

Cost of Accidents: Computation of costs, Utility of cost data

Plant Safety Inspection: Types, Inspection procedure, Safety sampling techniques, Job safety analysis (JSA), Safety surveys, Safety audits

CO4	**Conduct accident investigations and analyze safety performance metrics.**		
M4.01	Describe the key principles and processes involved in accident investigation.	3	Understanding
M4.02	Utilize various tools and techniques for data collection during investigations.	3	Understanding
M4.03	Analyze accidents using different analytical methods, including MORT and Change Analysis.	2	Understanding
M4.04	Conduct case studies to apply investigation techniques and improve safety protocols.	3	Applying
	Series Test – II	1	

Contents:

Accident Investigation: Why? When? Where? Who? & How?, Basics : Man, Environment & Systems

Process of Investigation: Tools, Data collection, Handling witnesses, Case study

Accident Analysis: Analytical techniques, System safety, Change analysis, MORT, Multi-events sequencing, TOR

Text / Reference

T/R	Author/ Book Title
T1	1.National Safety Council, Accident prevention manual for industrial operations. Chicago.(1982).
T2	2. Krishnan, N.V. Safety management in Industry. Jaico Publishing House, New Delhi.(1997).
R1	David L. Goetsch, Occupational Safety and health, PrenticeHall, 2010
R2	Ted S. Ferry, Modern Accident Investigation and Analysis, John Wiley &Sons, 2007
R3	Willie Hammer, Occupational Safety Management and Engineering, PrenticeHall, 1989
R4	Akhil Kumar Das, Industrial Safety Management, PHI,2020